

MTH441: Lab 9

Penalized GLMs

1. Using `glmnet` R package, fit a logistic regression model with lasso for the Titanic dataset, and compare the results with that of MLE estimation. Can you write down the final loss function for this estimator as well?
2. Fit a Poisson regression model to the Horseshoe Crab dataset here: https://search.r-project.org/CRAN/refmans/CatDataAnalysis/html/table_4.3.html, and understand the summary table.
3. Can you explain why a negative binomial model might be better for this dataset? Go ahead and fit this model and compare the likelihood values with 2.